VENKATKUMAR RAJAN

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EXPERIENCE

Generative AI Engineer — **Dec 2023 – Jul 2026**
EXL Service (India) Pvt. Ltd. — *Noida, India (Remote)*

- **AI-Powered Medical Credentialing Platform:** Designed and developed a reusable Python-based backend platform using Azure OpenAI (GPT-5), Azure AI Document Intelligence, Azure AI Search, and Retrieval-Augmented Generation (RAG) to automate extraction of **80+ provider credential fields** from complex healthcare documents. Built configurable extraction workflows with automated validation and data lifecycle management, achieving **70%+ extraction accuracy**.
- **Enterprise Document Intelligence Platform:** Designed and developed a modular, reusable Python backend platform for intelligent document processing across enterprise healthcare documents, including **CMS-1500, invoices, Explanation of Benefits (EOB), scheduling documents, and provider credentialing records**. Built a configuration-driven architecture using Azure OpenAI, Azure AI Document Intelligence, OCR, NLP, and Computer Vision, enabling rapid onboarding of new document workflows with minimal engineering effort. Integrated Azure Blob Storage, Azure Key Vault, SQL logging, and Azure DevOps CI/CD for scalable production deployment.
- **AI Document Classification System:** Developed a scalable document classification solution using Azure OpenAI capable of classifying **15+ enterprise document categories** with **97%+ accuracy**. Designed a configuration-driven architecture allowing new document categories to be onboarded without prompt redesign or code changes.
- **Enterprise AI Assistant Platform:** Developed configurable conversational AI assistants on Databricks by integrating Unity Catalog, enterprise knowledge bases, and Retrieval-Augmented Generation (RAG), enabling domain-specific AI assistants for business analytics, document intelligence, and healthcare workflows with integrated LLM evaluation and feedback learning.
- **Generative AI Solution:** Developed multiple enterprise Generative AI proof-of-concepts and internal AI productivity solutions, delivered technical demonstrations to prospective clients, and collaborated with cross-functional teams to accelerate AI adoption across U.S. healthcare and insurance business workflows.

Machine Learning Engineer — **Sept 2022 – Oct 2023**
AUGRAY — *Chennai, India*

- **Real-Time Ball Defect Detection:** Designed and deployed a real-time Computer Vision solution using MobileNetV2 on NVIDIA Jetson AGX for automated ball defect detection in manufacturing, achieving **90%+ accuracy**, reducing quality inspection errors by **40%**, and improving production throughput by **3×**.
- **Automated 3D Reconstruction Platform:** Designed and developed an end-to-end 3D reconstruction platform by integrating Photogrammetry and Neural Radiance Fields (NeRF), achieving **95% reconstruction accuracy** for non-reflective objects and **70%** for reflective surfaces while significantly reducing manual reconstruction effort.
- **Wall Segmentation and Color Visualization:** Developed semantic wall and floor segmentation models for automated interior visualization and color scheme generation, achieving **80% segmentation accuracy** and reducing manual design effort by **50+ hours per month**.
- **Virtual Shoe Try-On:** Designed and developed an AI-powered virtual shoe try-on solution using a dataset of **10,000+ annotated foot images**, achieving **79% placement accuracy** and contributing to a projected **15% reduction in customer return rates** during pilot evaluation.
- **Generative AI Solution:** Developed multiple Generative AI proof-of-concepts, including AI-powered marketing flyer generation, face texture synthesis, and conversational AI applications, delivering technical demonstrations to support client engagements & solution discovery.

TECHNICAL SKILLS

Programming	Python, C++, SQL
Frameworks	PyTorch, TensorFlow, OpenCV, MediaPipe, PyG, NetworkX, Gym, Librosa, PySpark, Open3D, PyTorch3D, GeoPandas, Rasterio, TorchGeo, LangChain, LangGraph, MCP, CrewAI, AutoGen
Frontend & Backend	React, FastAPI, Django, Node.js
Infrastructure	AWS, Azure, Databricks, Docker, Kubernetes, Kubeflow, MLflow
Edge Devices	NVIDIA Jetson AGX, Raspberry Pi
AI Specialization	Computer Vision, Spatial AI, Retrieval Systems, Graph Neural Networks, Reinforcement Learning, Edge AI

OPEN SOURCE & RESEARCH

- **Open Source (Ant Intelligence Ecosystem):** Seven **PyPI** libraries covering LLM evaluation, multimodal AI (vision, audio, document, time-series), and AI safety. **AntStudio** orchestrates all into a unified pipeline builder; **Arikizu OS** provides an edge-first AI runtime for local-first deployment. [Documentation]
- **Research & Credentials:** Publications in **IEEE, arXiv, and ResearchGate** on Generative AI, Computer Vision, and Retrieval Systems. **Kaggle Competition Expert**, 100+ competitions. Certified in **Self-Driving Cars, NVIDIA Jetson AI, and Microsoft Azure AI**.

EDUCATION

- **M.Tech. in Artificial Intelligence & Machine Learning** — **Rajasthan, India**
  Birla Institute of Technology and Science (BITS), Pilani — *Mar 2023 – Apr 2025*
- **B.E. in Electronics and Communication Engineering** — **Madurai, Tamil Nadu, India**
  SACS MAVMM Engineering College, Anna University — *Jul 2016 – Oct 2020*